

Statement of Teaching Philosophy

I believe that meaningful and effective learning occurs when students actively engage with course material and learn collaboratively in a welcoming and supportive classroom. I see my role as an instructor as encouraging students to ask questions, work with their peers, and stay curious about how course concepts connect to real-world contexts. In my teaching, I aim to prioritize a clear course structure and interactive learning strategies that help students succeed in the course while developing knowledge and skills they can carry into their future education and careers.

I put this philosophy into practice by creating opportunities for students to work through problems rather than simply receiving answers. As a teaching assistant for Organic Chemistry and Principles of Chemistry laboratories, I used activities such as safety scenarios and pre-lab concept discussions to help students prepare for experiments, identify important safety considerations, and connect laboratory procedures to the underlying chemistry. During lab, I asked students to work in small groups and circulated among the groups to check their progress, ask questions, and address misconceptions. Rather than immediately giving an answer when students encountered a problem, I encouraged them to explain their reasoning and consider possible next steps. This approach allowed me to see where students were struggling while helping them become more confident in solving problems independently. Student feedback reflected this approach; one student wrote, *"Sharear made frequent rounds to each and every student during lab, checking on their progress and ensuring they understood the process."*

My teaching and mentoring experiences have also shown me that students enter a course with different levels of preparation, prior knowledge, and confidence. I try to provide multiple ways for students to engage with a difficult concept, including diagrams, real-world examples, guided questions, and discussion with peers. I also value frequent, low-stakes opportunities for students to check their understanding before major assessments. In future courses, I plan to combine higher-stakes assessments with short quizzes, in-class polling, group problem solving, and other formative activities that provide regular feedback and help students recognize where they need additional practice. My research mentoring has strengthened this approach. I have worked with students from high school through graduate levels, including undergraduates who began with limited experience in computational chemistry and programming. By breaking projects into manageable steps, working through problems together, and gradually giving students greater independence, I helped one undergraduate complete a thesis and become a co-author on a peer-reviewed publication. My commitment to undergraduate mentoring was also recognized with an Exceptional Mentorship Award at the 5th Annual Research Symposium at Fayetteville State University.

I have continued to develop my teaching through formal pedagogical training and classroom experience. During graduate school, I completed the Certificate in University Teaching Skills program, where I received training in course design, student engagement, assessment, and instructional technology. This training encouraged me to think more deliberately about connecting course activities and assessments to specific learning goals. In Fall 2025, I served as a learning assistant for a general education course taught by my postdoctoral advisor, where I helped organize course materials, facilitated in-class and online group discussions, and occasionally served as a substitute lecturer. This experience expanded my preparation beyond laboratory instruction and provided additional opportunities to guide discussions, explain concepts to

students with different academic backgrounds and levels of preparation, and organize learning activities for a larger class.

In my future teaching, I would like to continue developing structured peer-learning activities that give students responsibility for explaining their reasoning to one another. For introductory courses and classes that include students with limited prior chemistry experience, I am particularly interested in approaches inspired by Peer-Led Team Learning (PLTL), where students learn by discussing problems, explaining their reasoning to peers, and working toward solutions together. I plan to incorporate more in-class problem solving and group activities that give students regular opportunities to practice reasoning independently, while also helping them use generative AI as a learning tool to create resources such as practice questions, flashcards, summaries, and other study materials that support their understanding of course concepts.

My teaching, mentoring, and research experiences prepare me to contribute across the chemistry curriculum, with particular strengths in biochemistry, biophysical chemistry, and computational approaches to biomolecular science. I am prepared to contribute to introductory and general chemistry courses and laboratories, as well as courses in biochemistry and biophysical chemistry. Building on my research in RNA structure and computational chemistry, I would also be excited to develop an upper-level course on biophysical and computational approaches to investigate biomolecular structure and function. In such a course, students could work with authentic biochemical and structural datasets, learn to interpret experimental measurements, and use tools such as Python or R for data analysis and visualization.

Across my courses, my goal is to help students build a strong foundation in chemistry and biochemistry while developing the confidence to ask questions, work through unfamiliar problems, and apply what they learn beyond the classroom. I hope to create learning experiences that prepare students not only for their next course, but also for the future academic and professional challenges.